전산화학 (2013년 1학기) 예제 및 과제 풀이

1. 강의노트 27 쪽, **예제 2**

(a) 압력과 온도에 대한 내부에너지 $U(P,T)$의 전미분은 다음과 같이 구할 수 있다.

$$dU = \Big(\frac{\partial U}{\partial P}\Big)_T dP + \Big(\frac{\partial U}{\partial T}\Big)_P dT. \tag{1}$$

(b) 부피와 온도에 대한 엔트로피 $U(V,T)$의 전미분은 다음과 같이 구할 수 있다.

$$dU = \Big(\frac{\partial U}{\partial V}\Big)_T dV + \Big(\frac{\partial U}{\partial T}\Big)_V dT. \tag{2}$$

(c) 압력과 온도에 대한 부피 $V(P,T)$의 전미분은 다음과 같이 구할 수 있다.

$$dV = \Big(\frac{\partial V}{\partial P}\Big)_T dP + \Big(\frac{\partial V}{\partial T}\Big)_P dT. \tag{3}$$

(d) (c)의 결과를 (b)에 넣에 정리하면

$$\begin{aligned} dU &= \Big(\frac{\partial U}{\partial V}\Big)_T \Big[\Big(\frac{\partial V}{\partial P}\Big)_T dP + \Big(\frac{\partial V}{\partial T}\Big)_P dT\Big] + \Big(\frac{\partial U}{\partial T}\Big)_V dT \\ &= \Big(\frac{\partial U}{\partial V}\Big)_T \Big(\frac{\partial V}{\partial P}\Big)_T dP + \Big(\frac{\partial U}{\partial V}\Big)_T \Big(\frac{\partial V}{\partial T}\Big)_P dT + \Big(\frac{\partial U}{\partial T}\Big)_V dT \\ &= \Big(\frac{\partial U}{\partial V}\Big)_T \Big(\frac{\partial V}{\partial P}\Big)_T dP + \Big[\Big(\frac{\partial U}{\partial V}\Big)_T \Big(\frac{\partial V}{\partial T}\Big)_P + \Big(\frac{\partial U}{\partial T}\Big)_V\Big] dT \end{aligned} \tag{4}$$

(4) 식을 (1) 식과 비교하면

$$\Big(\frac{\partial U}{\partial P}\Big)_T = \Big(\frac{\partial U}{\partial V}\Big)_T \Big(\frac{\partial V}{\partial P}\Big)_T$$

그리고

$$\Big(\frac{\partial U}{\partial T}\Big)_P = \Big(\frac{\partial U}{\partial V}\Big)_T \Big(\frac{\partial V}{\partial T}\Big)_P + \Big(\frac{\partial U}{\partial T}\Big)_V$$

2. 강의노트 27 쪽, **과제 1**

$$x = r\cos\phi, \quad y = r\sin\phi \Rightarrow r^2 = x^2 + y^2, \quad \tan\phi = \frac{y}{x}. \tag{5}$$

연쇄법칙을 이용하면,

$$\Big(\frac{\partial}{\partial x}\Big)_y = \Big(\frac{\partial r}{\partial x}\Big)_y \Big(\frac{\partial}{\partial r}\Big)_\phi + \Big(\frac{\partial \phi}{\partial x}\Big)_y \Big(\frac{\partial}{\partial \phi}\Big)_r, \tag{6a}$$

$$\Big(\frac{\partial}{\partial y}\Big)_x = \Big(\frac{\partial r}{\partial y}\Big)_x \Big(\frac{\partial}{\partial r}\Big)_\phi + \Big(\frac{\partial \phi}{\partial y}\Big)_x \Big(\frac{\partial}{\partial \phi}\Big)_r, \tag{6b}$$

이고,

$$\frac{\partial r}{\partial x} = \frac{x}{r} = \cos\phi, \quad \frac{\partial \phi}{\partial x} = -\frac{\sin\phi}{r}, \tag{7a}$$

$$\frac{\partial r}{\partial y} = \frac{y}{r} = \sin\phi, \quad \frac{\partial \phi}{\partial y} = \frac{\cos\phi}{r}, \tag{7b}$$

이므로,

$$\boxed{\begin{aligned} \frac{\partial}{\partial x} &= \cos\phi\frac{\partial}{\partial r} - \frac{\sin\phi}{r}\frac{\partial}{\partial \phi}, \quad (8a) \\ \frac{\partial}{\partial y} &= \sin\phi\frac{\partial}{\partial r} + \frac{\cos\phi}{r}\frac{\partial}{\partial \phi}. \quad (8b) \end{aligned}}$$

한편, g를 다음과 같이 정의하면,

$$g \equiv \frac{\partial}{\partial x} = \cos\phi\frac{\partial}{\partial r} - \frac{\sin\phi}{r}\frac{\partial}{\partial \phi}, \tag{9}$$

$\partial^2/\partial x^2$는 다음과 같이 쓸 수 있다.

$$\begin{aligned} \frac{\partial^2}{\partial x^2} &= \frac{\partial}{\partial x}\cdot\frac{\partial}{\partial x}, \\ &= \frac{\partial g}{\partial x}, \\ &= \frac{\partial r}{\partial x}\frac{\partial g}{\partial r} + \frac{\partial \phi}{\partial x}\frac{\partial g}{\partial \phi}, \qquad (\text{(6a) 식으로부터}) \\ &= \cos\phi\frac{\partial g}{\partial r} - \frac{\sin\phi}{r}\frac{\partial g}{\partial \phi}, \qquad (\text{(7a) 식과 (7b) 식으로부터}) \\ &= \cos\phi\frac{\partial}{\partial r}\left(\cos\phi\frac{\partial}{\partial r} - \frac{\sin\phi}{r}\frac{\partial}{\partial \phi}\right) - \frac{\sin\phi}{r}\frac{\partial}{\partial \phi}\left(\cos\phi\frac{\partial}{\partial r} - \frac{\sin\phi}{r}\frac{\partial}{\partial \phi}\right), \\ &= \cos^2\phi\frac{\partial^2}{\partial r^2} + \frac{\sin\phi\cos\phi}{r^2}\frac{\partial}{\partial \phi} - \frac{\sin\phi\cos\phi}{r}\frac{\partial^2}{\partial r\partial \phi} \\ &\quad + \frac{\sin^2\phi}{r}\frac{\partial}{\partial r} - \frac{\sin\phi\cos\phi}{r}\frac{\partial^2}{\partial \phi\partial r} + \frac{\sin\phi\cos\phi}{r^2}\frac{\partial}{\partial \phi} + \frac{\sin^2\phi}{r^2}\frac{\partial^2}{\partial \phi^2}. \\ &= \boxed{\cos^2\phi\frac{\partial^2}{\partial r^2} + \frac{2\sin\phi\cos\phi}{r^2}\frac{\partial}{\partial \phi} - \frac{2\sin\phi\cos\phi}{r}\frac{\partial^2}{\partial r\partial \phi} + \frac{\sin^2\phi}{r}\frac{\partial}{\partial r} + \frac{\sin^2\phi}{r^2}\frac{\partial^2}{\partial \phi^2}.} \end{aligned} \tag{10}$$

같은 방식으로 h를 다음과 같이 정의하면

$$h \equiv \frac{\partial}{\partial y} = \sin\phi\frac{\partial}{\partial r} + \frac{\cos\phi}{r}\frac{\partial}{\partial \phi}, \tag{11}$$

$\partial^2/\partial y^2$ 는 다음과 같이 쓸 수 있다.

$$\begin{aligned}
\frac{\partial^2}{\partial y^2} &= \frac{\partial}{\partial y}\cdot\frac{\partial}{\partial y}, \\
&= \frac{\partial h}{\partial y}, \\
&= \frac{\partial r}{\partial y}\frac{\partial h}{\partial r} + \frac{\partial \phi}{\partial y}\frac{\partial h}{\partial \phi}, \qquad \text{(Eq. (6b) 식으로부터)} \\
&= \sin\phi\frac{\partial h}{\partial r} + \frac{\cos\phi}{r}\frac{\partial h}{\partial \phi}, \qquad \text{((7a) 식과 (7b) 식으로부터)} \\
&= \sin\phi\frac{\partial}{\partial r}\left(\sin\phi\frac{\partial}{\partial r} + \frac{\cos\phi}{r}\frac{\partial}{\partial\phi}\right) + \frac{\cos\phi}{r}\frac{\partial}{\partial\phi}\left(\sin\phi\frac{\partial}{\partial r} + \frac{\cos\phi}{r}\frac{\partial}{\partial\phi}\right), \\
&= \sin^2\phi\frac{\partial^2}{\partial r^2} - \frac{\sin\phi\cos\phi}{r^2}\frac{\partial}{\partial\phi} + \frac{\cos\phi\sin\phi}{r}\frac{\partial^2}{\partial r\partial\phi} \\
&\quad + \frac{\cos^2\phi}{r}\frac{\partial}{\partial r} + \frac{\cos\phi\sin\phi}{r}\frac{\partial^2}{\partial\phi\partial r} - \frac{\cos\phi\sin\phi}{r^2}\frac{\partial}{\partial\phi} + \frac{\cos^2\phi}{r^2}\frac{\partial^2}{\partial\phi^2}. \\
&= \boxed{\sin^2\phi\frac{\partial^2}{\partial r^2} - \frac{2\cos\phi\sin\phi}{r^2}\frac{\partial}{\partial\phi} + \frac{2\cos\phi\sin\phi}{r}\frac{\partial^2}{\partial r\partial\phi} + \frac{\cos^2\phi}{r}\frac{\partial}{\partial r} + \frac{\cos^2\phi}{r^2}\frac{\partial^2}{\partial\phi^2}.}
\end{aligned} \tag{12}$$

(10) 식과 (12) 식으로부터, 다음과 같은 결과를 얻는다.

$$\begin{aligned}
\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} &= \cos^2\phi\frac{\partial^2}{\partial r^2} + \frac{2\sin\phi\cos\phi}{r^2}\frac{\partial}{\partial\phi} - \frac{2\sin\phi\cos\phi}{r}\frac{\partial^2}{\partial r\partial\phi} + \frac{\sin^2\phi}{r}\frac{\partial}{\partial r} + \frac{\sin^2\phi}{r^2}\frac{\partial^2}{\partial\phi^2} \\
&\quad + \sin^2\phi\frac{\partial^2}{\partial r^2} - \frac{2\cos\phi\sin\phi}{r^2}\frac{\partial}{\partial\phi} + \frac{2\cos\phi\sin\phi}{r}\frac{\partial^2}{\partial r\partial\phi} + \frac{\cos^2\phi}{r}\frac{\partial}{\partial r} + \frac{\cos^2\phi}{r^2}\frac{\partial^2}{\partial\phi^2}, \\
&= (\sin^2\phi + \cos^2\phi)\frac{\partial^2}{\partial r^2} + \frac{(\sin^2\phi + \cos^2\phi)}{r}\frac{\partial}{\partial r} + \frac{(\sin^2\phi + \cos^2\phi)}{r^2}\frac{\partial^2}{\partial\phi^2}, \\
&= \boxed{\frac{\partial^2}{\partial r^2} + \frac{\partial}{r\partial r} + \frac{\partial^2}{r^2\partial\phi^2}.}
\end{aligned} \tag{13}$$

3. 강의노트 28 쪽, **과제 2**

(a) X 와 x 가 다음과 같이 정의되므로

$$X = \frac{m_p x_p + m_e x_e}{m_p + m_e}, \tag{14}$$

$$x = x_e - x_p, \tag{15}$$

일차 편도함수는 다음과 같이 얻어진다.

$$\left(\frac{\partial X}{\partial x_p}\right)_{x_e} = \frac{m_p}{m_p + m_e}, \tag{16}$$

$$\left(\frac{\partial x}{\partial x_p}\right)_{x_e} = -1, \tag{17}$$

$$\left(\frac{\partial X}{\partial x_e}\right)_{x_p} = \frac{m_e}{m_e + m_e}, \tag{18}$$

$$\left(\frac{\partial x}{\partial x_e}\right)_{x_p} = 1. \tag{19}$$

연쇄법칙을 이용하면 다음과 같은 결과를 얻는다.

$$\begin{aligned} \frac{\partial}{\partial x_p} &= \frac{\partial X}{\partial x_p}\frac{\partial}{\partial X} + \frac{\partial x}{\partial x_p}\frac{\partial}{\partial x}, & (20)\\ &= \frac{m_p}{m_p + m_e}\frac{\partial}{\partial X} - \frac{\partial}{\partial x}, & (21) \end{aligned}$$

그리고,

$$\begin{aligned} \frac{\partial}{\partial x_e} &= \frac{\partial X}{\partial x_e}\frac{\partial}{\partial X} + \frac{\partial x}{\partial x_e}\frac{\partial}{\partial x}, & (22)\\ &= \frac{m_e}{m_e + m_e}\frac{\partial}{\partial X} + \frac{\partial}{\partial x}. & (23) \end{aligned}$$

(b) 같은 방법으로, Y 와 y 에 대해 다음과 같이 나타낼 수 있고,

$$Y = \frac{m_p y_p + m_e y_e}{m_p + m_e}, \tag{24}$$

$$y = y_e - y_p, \tag{25}$$

일차 편도함수는 다음과 같이 얻어진다.

$$\left(\frac{\partial Y}{\partial y_p}\right)_{y_e} = \frac{m_p}{m_p + m_e}, \tag{26}$$

$$\left(\frac{\partial y}{\partial y_p}\right)_{y_e} = -1, \tag{27}$$

$$\left(\frac{\partial Y}{\partial y_e}\right)_{y_p} = \frac{m_e}{m_e + m_e}, \tag{28}$$

$$\left(\frac{\partial y}{\partial y_e}\right)_{y_p} = 1. \tag{29}$$

연쇄법칙을 이용하면 다음과 같은 결과를 얻는다.

$$\begin{aligned} \frac{\partial}{\partial y_p} &= \frac{\partial Y}{\partial y_p}\frac{\partial}{\partial Y} + \frac{\partial y}{\partial y_p}\frac{\partial}{\partial y}, \\ &= \frac{m_p}{m_p + m_e}\frac{\partial}{\partial Y} - \frac{\partial}{\partial y}, & (30) \end{aligned}$$

그리고

$$\begin{aligned} \frac{\partial}{\partial y_e} &= \frac{\partial Y}{\partial y_e}\frac{\partial}{\partial Y} + \frac{\partial y}{\partial y_e}\frac{\partial}{\partial y}, \\ &= \frac{m_e}{m_e + m_e}\frac{\partial}{\partial Y} + \frac{\partial}{\partial y}. & (31) \end{aligned}$$

(c) (21) 식으로부터,

$$\frac{\partial^2}{\partial x_e^2} = \frac{\partial}{\partial x_e} \cdot \frac{\partial}{\partial x_e},$$
$$= \frac{\partial}{\partial x_e}\left(\frac{m_e}{m_e + m_p}\frac{\partial}{\partial X} + \frac{\partial}{\partial x}\right). \tag{32}$$

g를 다음과 같이 정의한다면,

$$g \equiv \frac{m_e}{m_e + m_p}\frac{\partial}{\partial X} + \frac{\partial}{\partial x}, \tag{33}$$

(32) 식은 다음과 같이 쓸 수 있다.

$$\begin{aligned}
\frac{\partial^2}{\partial x_e^2} &= \frac{\partial}{\partial x_e} \cdot \frac{\partial}{\partial x_e}, \\
&= \frac{\partial}{\partial x_e}\left(\frac{m_e}{m_e + m_p}\frac{\partial}{\partial X} + \frac{\partial}{\partial x}\right), \\
&= \frac{\partial g}{\partial x_e}, \\
&= \frac{\partial X}{\partial x_e}\frac{\partial g}{\partial X} + \frac{\partial x}{\partial x_e}\frac{\partial g}{\partial x}, \qquad \text{(from Eq. (20))} \\
&= \frac{m_e}{m_e + m_p}\frac{\partial g}{\partial X} + \frac{\partial g}{\partial x}, \qquad \text{(from Eqs. (16) and (17))} \\
&= \frac{m_e}{m_e + m_p}\frac{\partial}{\partial X}\left(\frac{m_e}{m_e + m_p}\frac{\partial}{\partial X} + \frac{\partial}{\partial x}\right) + \frac{\partial}{\partial x}\left(\frac{m_e}{m_e + m_p}\frac{\partial}{\partial X} + \frac{\partial}{\partial x}\right), \\
&= \boxed{\frac{m_e^2}{(m_e + m_p)^2}\frac{\partial^2}{\partial X^2} + \frac{2m_e}{m_e + m_p}\frac{\partial^2}{\partial X \partial x} + \frac{\partial^2}{\partial x^2}.}
\end{aligned} \tag{34}$$

같은 방법으로 (23) 식으로부터,

$$\frac{\partial^2}{\partial x_p^2} = \frac{\partial}{\partial x_p} \cdot \frac{\partial}{\partial x_p},$$
$$= \frac{\partial}{\partial x_p}\left(\frac{m_p}{m_e + m_p}\frac{\partial}{\partial X} - \frac{\partial}{\partial x}\right). \tag{35}$$

h를 다음과 같이 정의한다면,

$$h \equiv \frac{m_p}{m_e + m_p}\frac{\partial}{\partial X} - \frac{\partial}{\partial x}, \tag{36}$$

(35) 식은 다음과 같0이 쓸 수 있다.

$$\begin{aligned}
\frac{\partial^2}{\partial x_p^2} =& \frac{\partial}{\partial x_p} \cdot \frac{\partial}{\partial x_p}, \\
=& \frac{\partial}{\partial x_p}\left(\frac{m_p}{m_e+m_p}\frac{\partial}{\partial X} - \frac{\partial}{\partial x}\right), \\
=& \frac{\partial h}{\partial x_p}, \\
=& \frac{\partial X}{\partial x_p}\frac{\partial h}{\partial X} + \frac{\partial x}{\partial x_p}\frac{\partial h}{\partial x}, \qquad \text{(from Eq. (22))} \\
=& \frac{m_p}{m_e+m_p}\frac{\partial h}{\partial X} + \frac{\partial h}{\partial x}, \qquad \text{(from Eqs. (18) and (19))} \\
=& \frac{m_p}{m_e+m_p}\frac{\partial}{\partial X}\left(\frac{m_p}{m_e+m_p}\frac{\partial}{\partial X} - \frac{\partial}{\partial x}\right) + \frac{\partial}{\partial x}\left(\frac{m_p}{m_e+m_p}\frac{\partial}{\partial X} - \frac{\partial}{\partial x}\right), \\
=& \boxed{\frac{m_p^2}{(m_e+m_p)^2}\frac{\partial^2}{\partial X^2} - \frac{2m_p}{m_e+m_p}\frac{\partial^2}{\partial X \partial x} + \frac{\partial^2}{\partial x^2}.}
\end{aligned} \tag{37}$$

(d) (c)와 정확히 같은 방식으로,

$$\boxed{\frac{\partial^2}{\partial y_e^2} = \frac{m_e^2}{(m_e+m_p)^2}\frac{\partial^2}{\partial Y^2} + \frac{2m_e}{m_e+m_p}\frac{\partial^2}{\partial Y \partial y} + \frac{\partial^2}{\partial y^2},} \tag{38}$$

그리고

$$\boxed{\frac{\partial^2}{\partial y_p^2} = \frac{m_p^2}{(m_e+m_p)^2}\frac{\partial^2}{\partial Y^2} - \frac{2m_p}{m_e+m_p}\frac{\partial^2}{\partial Y \partial y} + \frac{\partial^2}{\partial y^2}.} \tag{39}$$

(e) (34) 식, (37) 식, (38) 식, 그리고 (39) 식으로부터,

$$\begin{aligned}
&\frac{1}{m_e}\left(\frac{\partial^2}{\partial x_e^2} + \frac{\partial^2}{\partial y_e^2}\right) + \frac{1}{m_p}\left(\frac{\partial^2}{\partial x_p^2} + \frac{\partial^2}{\partial y_p^2}\right) \\
=& \frac{1}{m_e}\left[\frac{m_e^2}{(m_e+m_p)^2}\frac{\partial^2}{\partial X^2} + \frac{2m_e}{m_e+m_p}\frac{\partial^2}{\partial X \partial x} + \frac{\partial^2}{\partial x^2}\right. \\
&\left. + \frac{m_e^2}{(m_e+m_p)^2}\frac{\partial^2}{\partial Y^2} + \frac{2m_e}{m_e+m_p}\frac{\partial^2}{\partial Y \partial y} + \frac{\partial^2}{\partial y^2}\right] \\
&+ \frac{1}{m_p}\left[\frac{m_p^2}{(m_e+m_p)^2}\frac{\partial^2}{\partial X^2} - \frac{2m_p}{m_e+m_p}\frac{\partial^2}{\partial X \partial x} + \frac{\partial^2}{\partial x^2}\right. \\
&\left. + \frac{m_p^2}{(m_p+m_p)^2}\frac{\partial^2}{\partial Y^2} - \frac{2m_p}{m_p+m_p}\frac{\partial^2}{\partial Y \partial y} + \frac{\partial^2}{\partial y^2}\right], \\
=& \frac{m_e}{(m_e+m_p)^2}\frac{\partial^2}{\partial X^2} + \frac{2}{m_e+m_p}\frac{\partial^2}{\partial X \partial x} + \frac{1}{m_e}\frac{\partial^2}{\partial x^2} \\
&+ \frac{m_e}{(m_e+m_p)^2}\frac{\partial^2}{\partial Y^2} + \frac{2}{m_e+m_p}\frac{\partial^2}{\partial Y \partial y} + \frac{1}{m_e}\frac{\partial^2}{\partial y^2} \\
&+ \frac{m_p}{(m_e+m_p)^2}\frac{\partial^2}{\partial X^2} - \frac{2}{m_e+m_p}\frac{\partial^2}{\partial X \partial x} + \frac{1}{m_p}\frac{\partial^2}{\partial x^2} \\
&+ \frac{m_p}{(m_p+m_p)^2}\frac{\partial^2}{\partial Y^2} - \frac{2}{m_e+m_p}\frac{\partial^2}{\partial Y \partial y} + \frac{1}{m_p}\frac{\partial^2}{\partial y^2},
\end{aligned}$$

$$
\begin{aligned}
&= \frac{m_e}{(m_e+m_p)^2}\left(\frac{\partial^2}{\partial X^2}+\frac{\partial^2}{\partial Y^2}\right)+\frac{m_p}{(m_e+m_p)^2}\left(\frac{\partial^2}{\partial X^2}+\frac{\partial^2}{\partial Y^2}\right)\\
&\quad+\frac{1}{m_e}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)+\frac{1}{m_p}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right),\\
&= \boxed{\frac{1}{m_e+m_p}\left(\frac{\partial^2}{\partial X^2}+\frac{\partial^2}{\partial Y^2}\right)+\left(\frac{1}{m_e}+\frac{1}{m_p}\right)\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right).} \qquad (40)
\end{aligned}
$$

4. 강의노트 31 쪽, **예제** (b), (c)

(a) 수업시간에 풀었음..

(b) $f(x,y)=2x-x^2+2y^2-y^4$ 의 임계점을 (x^*,y^*) 라 하면, 임계점에서

$$
\begin{aligned}
f_x(x^*,y^*) &= 0 \Rightarrow 2-2x^*=0 \Rightarrow x^*=1\\
f_y(x^*,y^*) &= 0 \Rightarrow 4y^*-4y^{*3}=0 \Rightarrow 4y^*(1+y^*)(1-y^*)=0\\
&\Rightarrow y^*=0 \quad \text{또는} \quad y^*=-1 \quad \text{또는} \quad y^*=1
\end{aligned}
$$

위 두 결과로부터 임계점은 $(0,0),\ (0,-1),\ (0,1)$ 이다.

Hessian 검사:

$$
\Delta_1 = f_{xx}(x^*,y^*) \quad \text{그리고} \quad \Delta_2 = f_{xx}(x^*,y^*)f_{yy}(x^*,y^*)-f_{xy}(x^*,y^*)^2.
$$

주어진 함수 $f(x,y)=2x-x^2+2y^2-y^4$ 에 대하여

$$
\begin{aligned}
f_{xx}(x,y) &= -2\\
f_{yy}(x,y) &= 4-12y^2\\
f_{xy}(x,y) &= 0
\end{aligned}
$$

i. $(0,0)$ 에서

$$
\begin{aligned}
f_{xx}(0,0) &= -2\\
f_{yy}(0,0) &= 4\\
f_{xy}(0,0) &= 0
\end{aligned}
$$

그러므로

$$
\Delta_1 = -2 < 0 \quad \text{and} \quad \Delta_2 = (-2)\times(4)-(0)^2 = -8 < 0
$$

$\Delta_2 < 0$ 이므로 $f(x,y)$ 는 $(0,0)$ 에서 안장점을 갖는다.

ii. $(0,-1)$ 에서

$$f_{xx}(0,-1) = -2$$
$$f_{yy}(0,-1) = -8$$
$$f_{xy}(0,-1) = 0$$

그러므로

$$\Delta_1 = -2 < 0 \quad \text{and} \quad \Delta_2 = (-2)\times(-8) - (0)^2 = 16 > 0$$

$\Delta_1 < 0$ 이고 $\Delta_2 > 0$ 이므로 $f(x,y)$ 는 $(0,1)$ 에서 극대값을 갖는다.

iii. $(0,1)$ 에서

$$f_{xx}(0,1) = -2$$
$$f_{yy}(0,1) = -8$$
$$f_{xy}(0,1) = 0$$

그러므로

$$\Delta_1 = -2 < 0 \quad \text{and} \quad \Delta_2 = (-2)\times(-8) - (0)^2 = 16 > 0$$

$\Delta_1 < 0$ 이고 $\Delta_2 > 0$ 이므로 $f(x,y)$ 는 $(0,-1)$ 에서 극대값을 갖는다.

(c) $f(x,y) = \frac{1}{2}x^2 - xy$ 의 임계점을 (x^*,y^*) 라 하면, 임계점에서

$$f_x(x^*,y^*) = 0 \Rightarrow x^* - y^* = 0 \Rightarrow x^* = y^*$$
$$f_y(x^*,y^*) = 0 \Rightarrow -x^* = 0 \Rightarrow x^* = 0$$

위 두 결과로부터 임계점은 $(0,0)$ 이다.

Hessian 검사:

$$\Delta_1 = f_{xx}(x^*,y^*) \quad \text{그리고} \quad \Delta_2 = f_{xx}(x^*,y^*)f_{yy}(x^*,y^*) - f_{xy}(x^*,y^*)^2.$$

주어진 함수 $f(x,y) = \frac{1}{2}x^2 - xy$ 에 대하여

$$f_{xx}(x,y) = 1$$
$$f_{yy}(x,y) = 0$$
$$f_{xy}(x,y) = -1$$

$(0,0)$ 에서

$$f_{xx}(0,0) = 1$$
$$f_{yy}(0,0) = 0$$
$$f_{xy}(0,0) = -1$$

그러므로

$$\Delta_1 = 1 > 0 \quad \text{and} \quad \Delta_2 = (1) \times (0) - (-1)^2 = -1 < 0$$

$\Delta_2 < 0$ 이므로 $f(x,y)$ 는 $(0,0)$ 에서 안장점을 갖는다.